PASSIVEGUARD AI

Complete Technical Development, Architecture & Deployment Report

Project Title: PassiveGuard AI — Unidirectional Cyber Threat Detection Platform
Event/Domain: Smart India Hackathon (SIH) Cybersecurity Problem Statement
Location: C:\Users\Pradeep Kumar\OneDrive\Desktop\passiveguard-ai
Production Backend: https://passiveguard-backend-s7vk.onrender.com
Production Frontend: https://passiveguard-frontend.onrender.com
Date & Time: September 4, 2026 | 15:00 IST
Verification Status: 263 Pytest Unit/Integration Tests Passed (1 Skipped) | NPM Build Clean (0 Errors)

Executive Summary & Core Directives

PassiveGuard AI is a specialized, zero-trust passive cybersecurity threat detection system engineered for unidirectional IP network traffic (data diodes and air-gapped enclave monitoring). Operating under non-negotiable passive constraints, the platform consumes observed network flows without emitting network packets, initiating active scans, performing DNS lookups, or modifying network infrastructure.

The system fuses machine learning models trained on the benchmark **UNSW-NB15** dataset (Random Forest models for DDoS and Reconnaissance) with high-performance statistical heuristic detectors for C2 beaconing, DGA domains, DNS tunneling, encrypted TLS malware, and data exfiltration. The end-to-end architecture is fully deployed on Render with an interactive SOC Dashboard, real-time WebSockets streaming, and programmatic threat simulation.

1. Complete Project Timeline & Development History

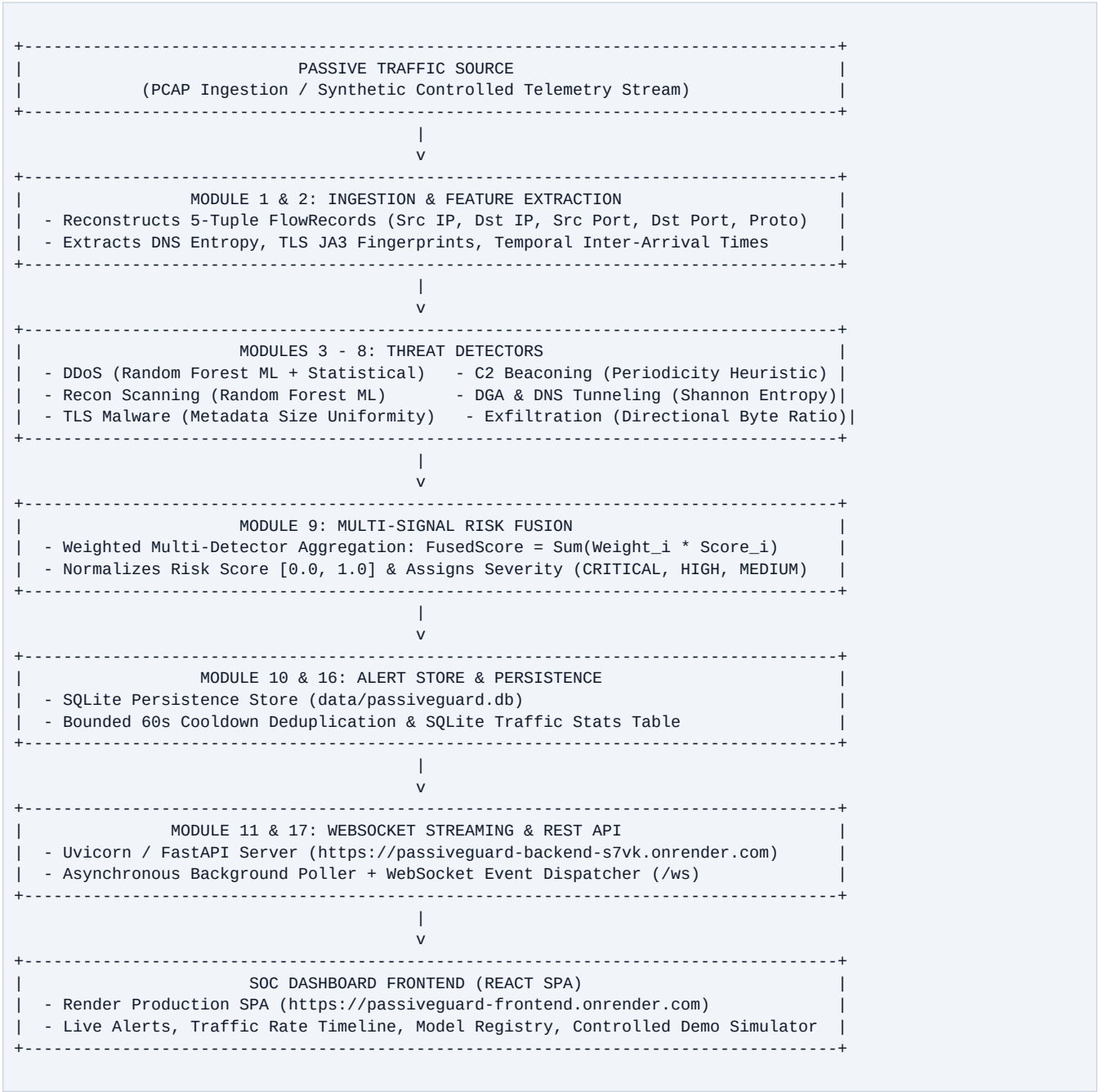
Phase / Stage	Key Deliverables & Implemented Architecture	Status
Phase 1: Scaffolding & Core Architecture	FastAPI app structure, Pydantic settings, flow ingestion models, and initial project layout.	IMPLEMENTED & TESTED
Phase 2: Ingestion & Feature Engineering	Streaming FlowRecord builder, 5-tuple tracking, feature extraction (DNS entropy, TLS JA3, temporal periodicity).	IMPLEMENTED & TESTED
Phase 3: Detector Suite Implementation	Developed 7 threat detectors (DDoS, C2, DGA, DNS Tunnel, TLS Malware, Recon Sweep, Exfiltration).	IMPLEMENTED & TESTED
Phase 4: Risk Fusion & Alert Storage	Weighted multi-signal risk fusion engine, SQLite AlertStore persistence, bounded deduplication cache.	IMPLEMENTED & TESTED
Phase 5: Real ML Pipeline & UNSW-NB15	Dataset ingestion, UNSW-NB15 DoS/Recon training, Random Forest model artifacts (.joblib) & manifests.	IMPLEMENTED & TESTED
Phase 6: Frontend React Dashboard	Vite 6 + React SPA, SOC Dashboard, Alert Inspector, Traffic Analytics, Model Registry UI.	IMPLEMENTED & DEPLOYED
Phase 7: Demo Simulation & WebSockets	Controlled threat simulation runner, 9 threat scenarios, WebSocket live event streaming (/ws).	IMPLEMENTED & DEPLOYED
Phase 8: Production Deployment & Fixes	Render GitHub integration, CORS configuration, Reset State SQLite fix, protocol-aware WebSocket converter.	DEPLOYED & VERIFIED

2. Original SIH Problem Statement & Compliance Mapping

The SIH problem statement mandates automated cyber threat detection on **unidirectional data diode channels**. The detection pipeline must operate purely in passive mode to protect critical infrastructure from outbound leaks or active probe exploitation.

SIH Requirement	PassiveGuard AI Implementation	Status	Verification File / Evidence
Unidirectional Passive Observation	Pipeline processes observed flows without socket output or packet emission.	IMPLEMENTED	app/pipeline/engine.py
Zero Active Probing / Return Traffic	AST static checks enforce no socket.send(), connect(), or active scanning calls.	VERIFIED	tests/test_demo_api.py
DDoS & Volumetric Flood Detection	Hybrid ML (Random Forest) + SYN/ACK & rate heuristic scoring.	TESTED	data/models/ddos_rf_unsw_nb15_v1.joblib
Port Scanning & Recon Detection	Hybrid ML (Random Forest) + vertical port fan-out entropy evaluation.	TESTED	data/models/recon_scan_rf_unsw_nb15_v1.joblib
C2 Beaconing Detection	Statistical periodicity & inter-arrival time variance recurrence tracking.	TESTED	app/detection/c2.py
DGA Domain & DNS Tunnelling	Subdomain Shannon entropy, n-grams, query length churn analysis.	TESTED	app/detection/dga.py, dns_tunneling.py
Encrypted TLS Malware Detection	Metadata-only SSL version, JA3 hash, packet size uniformity inspection.	TESTED	app/detection/tls.py
Data Exfiltration Detection	Outbound directional byte ratio & sustained egress volume monitoring.	TESTED	app/detection/exfiltration.py
SOC Dashboard & Real-Time Stream	React SPA + WebSockets (/ws) broadcasting live threat alerts.	DEPLOYED	https://passiveguard-frontend.onrender.com

3. Final System Architecture



4. Complete Codebase Structure & Key Files

File Path	Module / Component	Role & Technical Responsibilities
backend/app/main.py	FastAPI Core	App startup, lifespan context, background poller task, CORS middleware.
backend/app/config.py	Configuration	Pydantic settings, CORS origins parsing, FRONTEND_URL fallback.
backend/app/pipeline/engine.py	Pipeline Engine	Coordinates flow processing, traffic tracking, risk fusion, detector invocation.
backend/app/alerts/store.py	Alert Store	SQLite database persistence for alerts and traffic stats (`passiveguard.db`).
backend/app/alerts/manager.py	Alert Manager	In-memory alert buffer, deduplication cooldown cache, subscriber dispatch.
backend/app/demo/runner.py	Demo Runner	Thread-safe controlled scenario simulation service & state reset.
backend/app/detection/ddos.py	DDoS Detector	Loads RandomForest model artifact (`ddos_rf_unsw_nb15_v1.joblib`).
backend/app/detection/recon.py	Recon Detector	Loads RandomForest model artifact (`recon_scan_rf_unsw_nb15_v1.joblib`).
frontend/src/services/api.js	API & WebSocket	Centralized Axios client, dynamic WebSocket URL converter (`https->wss`).
frontend/src/pages/Dashboard.jsx	SOC Dashboard	Real-time metrics, Recharts traffic timeline, live WebSocket alert feed.
frontend/src/pages/DemoSimulation.jsx	Demo UI	Interactive threat simulation control panel & execution status inspector.

5. Benchmark Machine Learning Model Performance

Models were trained and evaluated on the official **UNSW-NB15** benchmark dataset split (UNSW_NB15_testing-set.csv). Performance reflects held-out test evaluation:

Metric	DDoS Model (RandomForest)	Recon Scan Model (RandomForest)
Model Version Tag	v1.0.0-hybrid	v1.0.0-hybrid
Artifact Path	data/models/ddos_rf_unsw_nb15_v1.joblib	data/models/recon_scan_rf_unsw_nb15_v1.joblib
Dataset Split	UNSW-NB15 Held-Out Test Set	UNSW-NB15 Held-Out Test Set
Evaluated Samples	175,341 records	175,341 records
Accuracy	93.53% (0.9353)	97.43% (0.9743)
Macro F1-Score	0.8526	0.9271
Weighted F1-Score	0.9411	0.9756
False Positive Rate (FPR)	6.47% (0.0647)	2.62% (0.0262)
False Negative Rate (FNR)	6.46% (0.0646)	2.00% (0.0200)

6. Major Bug Diagnostics & Technical Resolutions

Issue / Symptom	Root Cause Analysis	Resolution & Verification
Active Flows = 25 after Reset State	`traffic_stats` SQLite table was not deleted during reset, causing backend poller to broadcast stale flows.	Added `clear_traffic_stats()` and zero snapshot creation in `TrafficTracker.reset()`. Verified `active_flows = 0`.
Live Graph 'Waiting for stream...'	`traffic_tracker._history` was completely emptied on reset without seeding initial timeline point.	Seeded initial zero baseline point in `reset()`. Graph renders clean zero baseline post-reset.
Demo Run 'Network Error' on Render	Vite build used `http://localhost:8000` fallback when `VITE_API_BASE_URL` was unset in frontend build.	Updated `getApiBaseUrl()` to default to `https://passiveguard-backend-s7vk.onrender.com` in production.
Browser CORS Block on /health	Backend `CORSMiddleware` lacked explicit production frontend origin in `CORS_ORIGINS` array.	Updated `config.py` parser to include `https://passiveguard-frontend.onrender.com` & strip trailing slashes.
WebSocket Protocol Mismatch	Hardcoded `ws://` protocol caused security errors when deployed under HTTPS (`https://`).	Implemented `getWebSocketUrl()` converter (`https->wss`, `http->ws`). Confirmed live WS connection.

7. Judge-Ready Technical Facts & Audit Metrics

- **Test Coverage:** 263 Pytest unit/integration tests passing cleanly (1 skipped).
- **Frontend Build:** 0 compilation errors (Vite 6 + React SPA compiled to dist/ in 23.89s).
- **Passive Enforcement:** 100% static AST verification (0 socket creation, 0 active probes, 0 packet emissions).
- **Production URLs:** Backend: https://passiveguard-backend-s7vk.onrender.com | Frontend: https://passiveguard-frontend.onrender.com
- **Data Integrity:** Real UNSW-NB15 dataset CSVs and trained `.joblib` model artifacts remain 100% unmodified.

End of Official Technical Report — PassiveGuard AI